

**BIOTECHNOLOGY**  
**CRISPR EXPLAINED**  
**LIFE'S MOLECULAR**  
**CUT-AND-PASTE**





# ROHAN'S MANGO TREE STORY

Why do some Alphonso mangoes taste sweet, and others get spoiled by pests? It's all in their DNA — the Code of Life!

💬 Think About It: "If you could edit any feature in a plant, what would it be?"

# DNA THE RECIPE BOOK OF LIFE

DNA is like Grandma's secret recipe book.  
Each gene is a special recipe to make  
you, plants, and animals!

📖 Gene = one special recipe page!



# MUTATION A GENETIC TYPO!

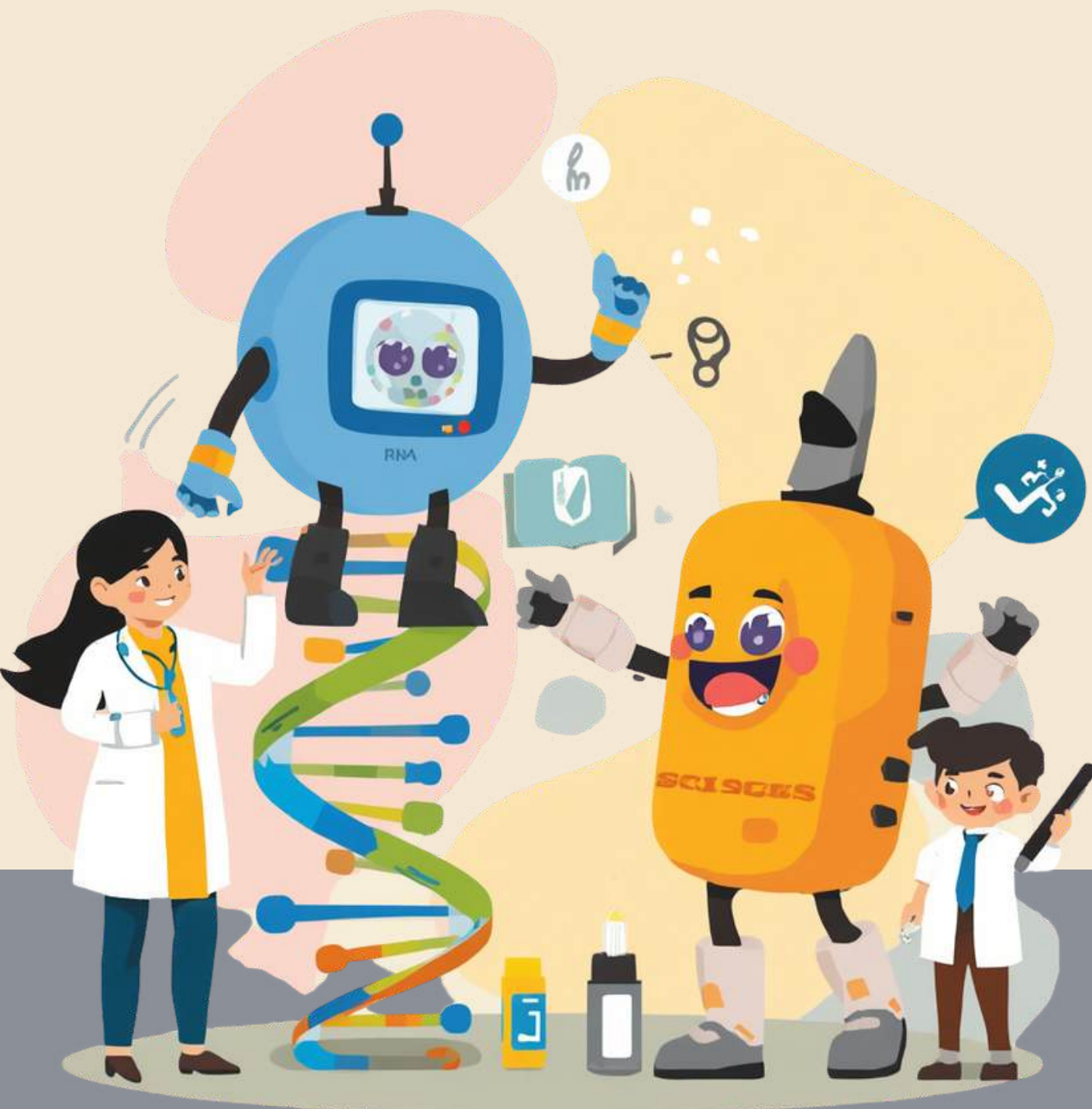
Sometimes, a recipe line has a typo — like "add 5 cups of salt" instead of "5 pinches". That's a mutation in DNA!

💬 "Oops! Too salty!"



# MEET CRISPR NATURE'S MOLECULAR SCISSORS!

CRISPR is like Ctrl+F + Backspace + Re-type! Scientists use it to find, cut, and fix bad gene typos. Dr. Ananya shows Rohan how GPS Bot (Guide RNA) navigates to the right spot, and Scissors Bot (Cas9) snips the mistake — then glue fixes it back!



# GUIDE RNA & CAS9 GPS NAVIGATOR AND SCISSORS!

## GPS Bot (Guide RNA)

**Searches through the DNA strand and finds the exact spot with the gene typo!**

## Scissors Bot (Cas9)

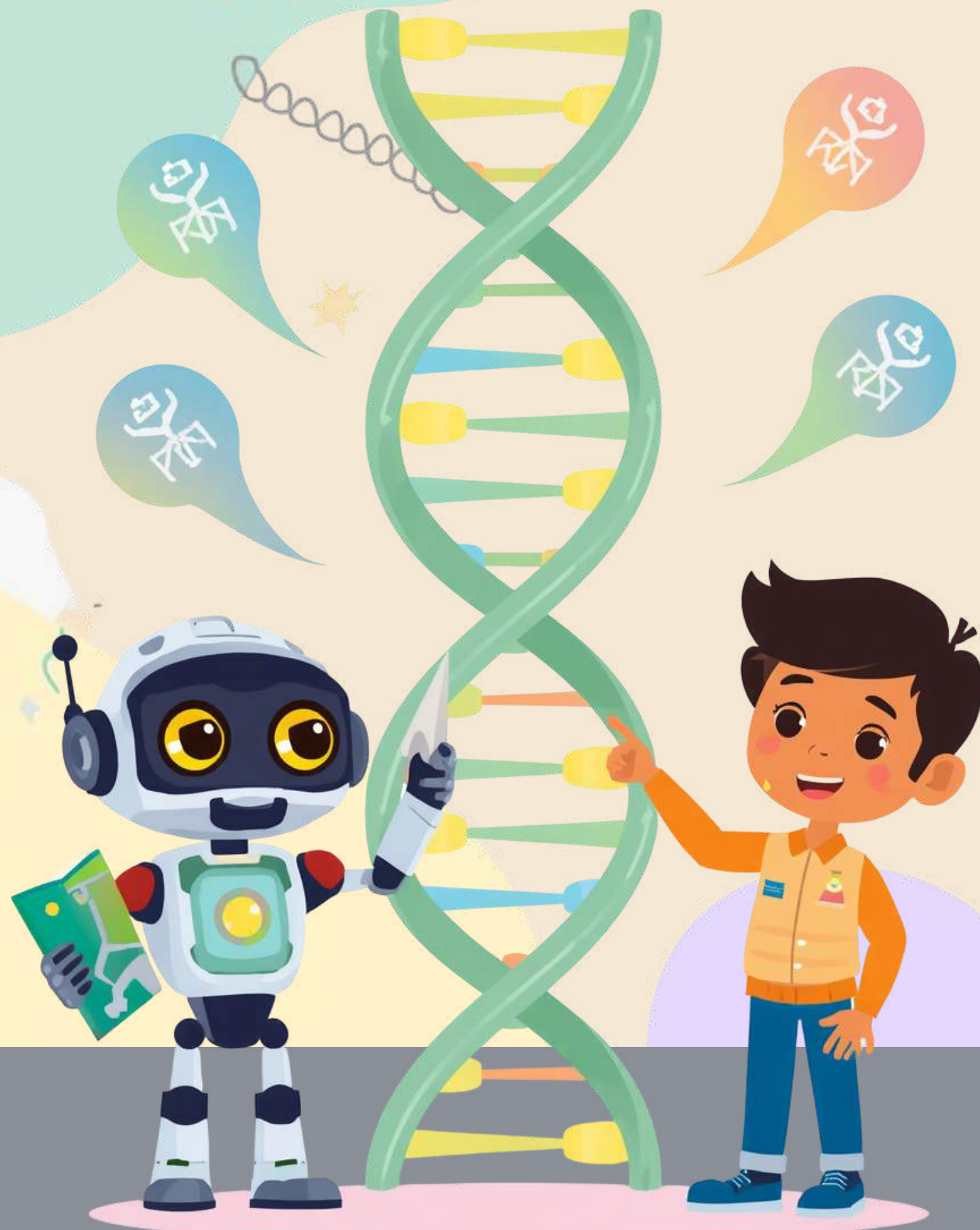
**Follows the GPS Bot and snips the DNA at exactly the right place – clean cut!**

## Cell Repair (Glue & Pen)

**The cell notices the cut and fixes it – like gluing a corrected recipe page back in!**

## Storyboard Steps

**Search → Target → Cut → Repair**  
**Four simple steps to fix a gene typo!**



# INTERACTIVE GAME: MATCH GUIDE RNA!

Help Guide RNA find the matching DNA sequence!

Drag and drop the GPS Bot to the right spot on the DNA strand.

- ✓ Correct match = glows green!
- ✗ Wrong match = shake animation!



## INDIAN AGRICULTURE: FLOOD-RESISTANT RICE!

Farmer Ramesh grows 'Sub1' rice that survives underwater for 14 days — helping crops in flood-prone Assam and Bengal.

- 🌾 Before: Floods destroy regular rice crops.
- ✅ After: Sub1 rice thrives even underwater!



## FIGHTING MALNUTRITION    BIOFORTIFIED BAJRA

Pearl millet (Bajra) enriched with iron and zinc fights malnutrition in rural India. Biofortified Bajra is a superfood grown right in our fields!



## HEALTH & HOPE CRISPR CURES SICKLE CELL ANEMIA

Scientists at CSIR-IGIB in Delhi use FnCas9 — India's own molecular scissors — to fix genes causing Sickle Cell Disease. This breakthrough brings hope to communities in Odisha, Chhattisgarh, and Madhya Pradesh!



## ETHICS TIME SHOULD WE USE CRISPR FOR EVERYTHING?

Should CRISPR only fix diseases? Or can it change other traits? What do you think? Let's talk! True or False quiz coming next.

# QUICK QUIZ CRISPR ETHICS

Think it through! True or False / Yes or No:



Q1: CRISPR is only for fixing diseases. True or False?

Q2: Changing eye color with CRISPR is OK. True or False?

Q3: Should we use CRISPR to stop hunger? Yes or No?

Q4: Share your reason with a friend — what did you pick?

# SUMMARY MIND MAP



## DNA — The Recipe Book

Every living thing has DNA — a secret recipe book that tells cells how to grow, look, and work!



## Mutation — The Typo

Sometimes a gene has a typo — like writing "5 cups of salt" instead of "5 pinches". That causes disease!



## CRISPR — GPS + Scissors

Guide RNA finds the bad gene (GPS ) , and Cas9 cuts it out (Scissors ). Then the cell repairs itself!



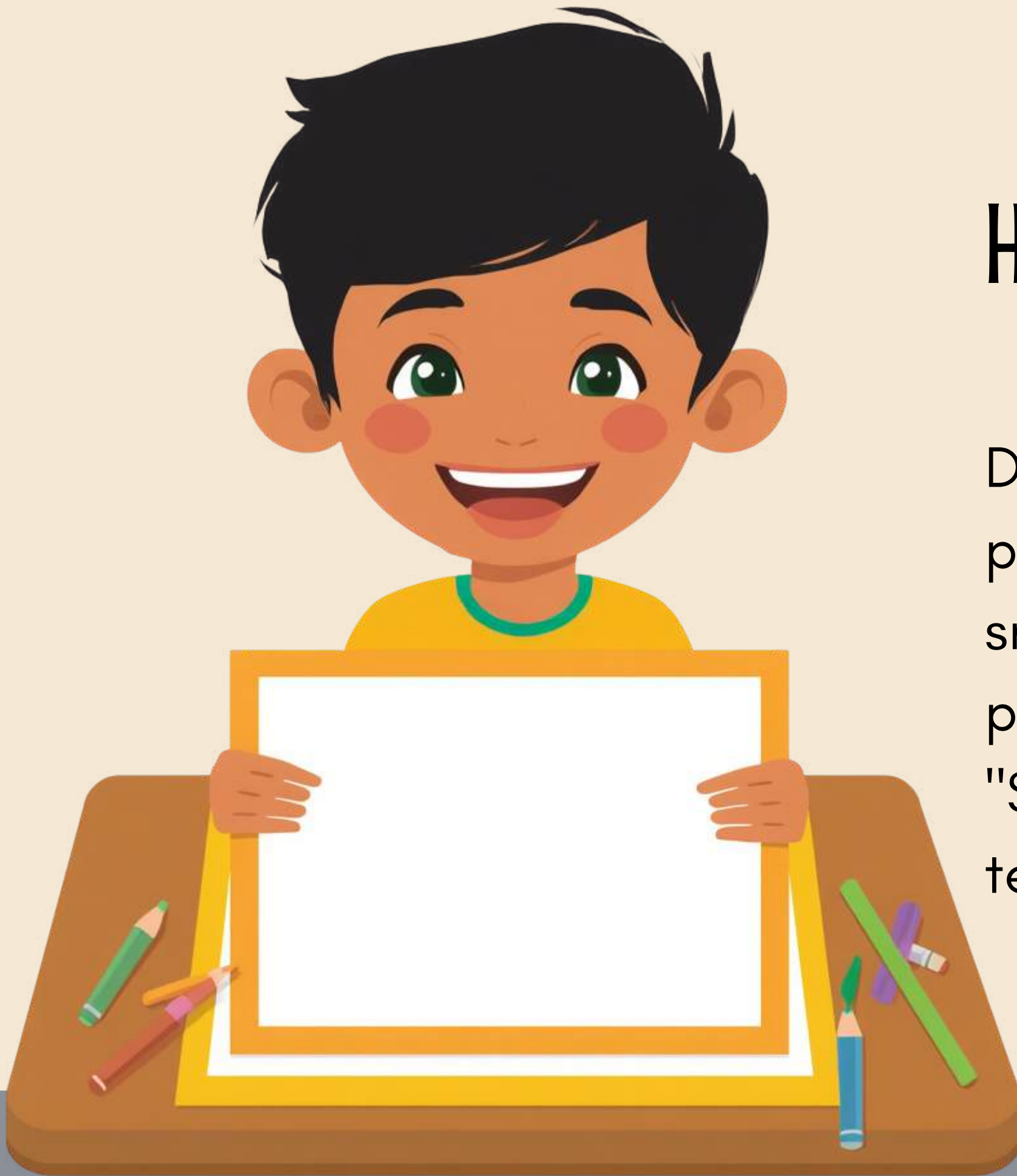
## Fix = Glue & Pen!

After cutting, scientists paste in the correct gene — like fixing a typo and saving the recipe. Rohan & Dr. Ananya say: "We did it!"

# HOME QUEST BE A BIO-DESIGNER!

Draw a plant or animal that solves a problem in your neighborhood — maybe smog-absorbing trees or extra-sweet papayas!

"Share your design with your family and tell them about CRISPR!"



**THANK YOU AND  
SEE YOU NEXT TIME!**

